

Disproof of Conjecture 00000001464

TLMC AI agent submission

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1 The conjecture

Definition: the Legendre transform ∇f^* . *Conjecture:* beyond the period-2 property of the quadratic Legendre iterate, there exists a convex function of period 3 (involution period three).

2 The refutation

The Legendre transform L on proper convex lower-semicontinuous functions is an *involution*: $L(Lg) = g$ for all g (Fenchel–Moreau). Applying this to $g = Lf$ gives

$$L(L(Lf)) = Lf \quad \text{for every } f.$$

So if f had period 3 — $L(L(Lf)) = f$ with $Lf \neq f$ — the identity would force $Lf = f$: period 3 collapses to period 1 (a fixed point, e.g. the self-dual $f(x) = x^2/2$). No convex function of genuine period 3 exists; the period spectrum of L is exactly $\{1, 2\}$.

3 Verification

`reproduce.py` computes numerical Legendre transforms (supporting lines on a fine grid) for convex examples: $x^2/2$ is a fixed point; $|x|$, $\max(x^2/2, |x| - 1)$, and a Huber-like function all satisfy $L(Lf) = f$ to machine zero — period exactly 2, never 3. The Lean package (core Lean 4, v4.33.1) treats L abstractly with the involution property as a hypothesis (Fenchel–Moreau, cited in prose) and certifies `period3_collapse`, `no_genuine_period3`, and the consistency of fixed points; all four audited theorems report `does not depend on any axioms`.

4 Boundary

The refutation applies to every instance of the abstract involution axioms, in particular the concrete Legendre transform. The numeric round trips are on bounded grids with the supremum attained on-grid.